WORKED SOLUTION

# STEP III 1992, Question 6

Pure • Trigonometric Reduction Formulae

**STEP III 1992, Question 6**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q6

## The key idea

Both reductions rest on turning a difference of trigonometric functions into a product that contains a factor $\sin x$, which then cancels the awkward denominator. For $I_n$ the tool is $\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$; for $J_n$ it is the sum-to-product formula for $\sin P - \sin Q$. Once the denominators have gone, what is left is an elementary integral, and the two reduction formulae telescope.

## Solution

First a word on the integrands. Each has $\sin x$ in the denominator, which vanishes at $x = 0$ and $x = \pi$, but the singularities are removable: since $\sin(nx)/\sin x \to n$ as $x \to 0$ and $\sin(nx)/\sin x \to \pm n$ as $x \to \pi$, both integrands stay bounded. So all the integrals below are ordinary integrals of continuous functions.

### The relation $I_n - I_{n-1} = J_n$

Using $\sin^2 A - \sin^2 B = (\sin A - \sin B)(\sin A + \sin B)$ and the sum-to-product formulae,

$$\sin^2 A - \sin^2 B = (2\cos\tfrac{A+B}{2}\sin\tfrac{A-B}{2})(2\sin\tfrac{A+B}{2}\cos\tfrac{A-B}{2}) = \sin(A+B)\sin(A-B).$$

With $A = nx$ and $B = (n-1)x$ this gives

$$\sin^2(nx) - \sin^2((n-1)x) = \sin((2n-1)x)\sin x.$$

Therefore, for $n \geq 1$,

$$I_n - I_{n-1} = \int_0^\pi \frac{x[\sin^2(nx) - \sin^2((n-1)x)]}{\sin^2 x}\,\mathrm{d}x = \int_0^\pi \frac{x\sin(2n-1)x}{\sin x}\,\mathrm{d}x = J_n,$$

as required. (For $n = 1$ we read $I_0$ as the integral of $x\sin^2 0/\sin^2 x = 0$, so $I_0 = 0$.)

### A reduction formula for $J_n$

Now compare $J_{n+1}$ with $J_n$. By the sum-to-product formula $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$,

$$\sin(2n+1)x - \sin(2n-1)x = 2\cos(2nx)\sin x,$$

so again the $\sin x$ cancels:

$$J_{n+1} - J_n = 2\int_0^\pi x\cos(2nx)\,\mathrm{d}x.$$

Integrating by parts with $u = x$ and $\mathrm{d}v = \cos(2nx)\,\mathrm{d}x$,

$$\begin{aligned}\int_0^\pi x\cos(2nx)\,\mathrm{d}x &= \Big[\frac{x\sin(2nx)}{2n}\Big]_0^\pi - \frac{1}{2n}\int_0^\pi \sin(2nx)\,\mathrm{d}x \\ &= 0 + \frac{1}{4n^2}\Big[\cos(2nx)\Big]_0^\pi = \frac{1}{4n^2}(1-1) = 0,\end{aligned}$$

since $\sin(2n\pi) = 0$ and $\cos(2n\pi) = 1$ for integer $n$. The reduction formula is therefore

$$J_{n+1} = J_n \qquad (n \geq 1).$$

**Ask yourself** A reduction formula that says $J_{n+1} = J_n$ looks suspiciously trivial. Is that really the answer?

Yes: the $2\cos(2nx)$ that appears integrates against $x$ to zero over a whole number of periods, so nothing changes from one $n$ to the next. To pin the value down we only need the first term:

$$J_1 = \int_0^\pi \frac{x\sin x}{\sin x}\,\mathrm{d}x = \int_0^\pi x\,\mathrm{d}x = \frac{\pi^2}{2},$$

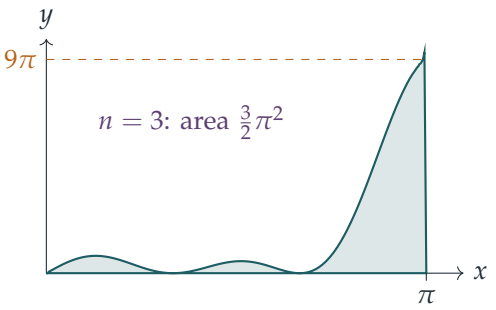
and hence $J_n = \frac{1}{2}\pi^2$ for every positive integer $n$.

### The area under $C$

Since $x \geq 0$ and $\sin^2(nx)/\sin^2 x \geq 0$ on $(0,\pi)$, the curve never dips below the $x$-axis, so the area under $C$ is exactly $I_n$. Summing the first relation (it telescopes) and using $I_0 = 0$,

$$I_n = I_0 + \sum_{k=1}^{n}(I_k - I_{k-1}) = \sum_{k=1}^{n} J_k = n\cdot\frac{\pi^2}{2}.$$

So the area under $C$ is $\frac{1}{2}n\pi^2$, as required.




## Checks

- $n = 1$: the curve is $y = x$, and the area of that triangle is $\frac{1}{2}\pi^2$. ✓
- Simpson's rule on the integrals for $n = 1, \ldots, 8$ gives $J_n \approx 4.9348022 = \frac{1}{2}\pi^2$ every time and, for example, $I_3 \approx 14.8044066 = \frac{3}{2}\pi^2$.
- The curve rises to $y = n^2\pi$ at $x = \pi$, yet the area grows only linearly in $n$: the peaks get taller but narrower (width about $\pi/n$), which is consistent.

## Common slips

- Forgetting that $I_0 = 0$ (or never checking what $I_0$ means), so the telescoping sum has no starting value.
- Sign errors in $\int x\cos(2nx)\,\mathrm{d}x$; it is cleaner to note that the boundary term vanishes and the remaining integral of $\sin(2nx)$ over $[0, \pi]$ is zero.

## Beyond STEP

The function $\sin^2(nx)/\sin^2 x$ is, up to a factor $1/n$, the *Fejér kernel* of Fourier analysis, and $\sin\big((2n-1)x\big)/\sin x$ is essentially the *Dirichlet kernel*. The fact you found here, that their integrals do not depend on $n$ while the functions concentrate at the ends of the interval, is exactly what makes them useful for proving that Fourier series converge. Stein and Shakarchi's *Fourier Analysis: An Introduction* develops this beautifully; for the real-analysis background, see Abbott, *Understanding Analysis*.

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